

CODE ANALYSIS — EXIT RECHECK (REV E) · P2-A-101 REV K / A-102 REV G · D-079

Supplements P2-G-002 Rev D (frozen, full analysis); only what the curved envelope, the moved stairs / exits and the ST-2 tower change is restated. Method and factors as Rev D. Not a code review: architect + AHJ confirm (D-008).

1 WHAT CHANGED (all ASSUMED)

- Envelope R = 20 ft corners (D-077); ST-3 slid west (X9 to x 183.7), ST-4 slid north (X10 to y 32), X1 to y 50; ST-2 is a 20 ft cylindrical tower, exit X6 on its east face (D-078).
- New 900 SF mech / elec bay (x 136-166), clear of EXIT (N) at X5. Level 2 as P2-A-102 Rev G (strip landing, rounded NE loop corner, D-079).
- Counts unchanged: E1 + X1-X11 = 12 openings, 4 stairs x 76 in, 2,200 seats.

2 OCCUPANT LOAD — REV D vs REV E (T1004.5 factors, Rev D method)

SPACE	REV D SF	REV E SF	D LOAD	E LOAD
Boys + girls locker (2)	7,014	7,014	142	142
Event lockers 1-4	3,600	3,600	72	72
Flex / team assembly	2,008	1,767	134	118
Flex / storage (SW)	975	972	4	4
Equipment storage + room	1,000	1,000	4	4
Mech / elec (6 rooms)	3,496	4,235	15	17
First aid	120	120	1	1
Concession	275	275	2	2
Chair / table / stage storage (annex)	2,100	2,100	7	7
Lower tier seats (telescopic)			1100	1100
L1 without the event floor			1481	1467
Event floor, chairs only (D-054)	16,416	16,416	2346	2346
L1 TOTAL, design case			3827	3813
L2 worst case (D-052)			1462	1460

Rev E changes: flex / team assembly -241 SF (ST-4 now fully inside it), MECH (NW / N / NE) lose area to the radius and, for MECH (N), to the tower (50 SF of it lies under the ST-2 tower: the Rev K room table did not deduct this), bay adds 3. L2: S&C 5,038 SF = 101, loop 5,006 SF = 101, rail strip 84; base 1275. Rev D method reproduced first: Rev J geometry gives 381 + 1,100 = 1,481 (matches).

3 LEVEL 1 EXIT CAPACITY (doors 0.15 in / occupant, 64 in per pair)

CHECK (in)	329	1,095	2,346	3,284	PROVIDED
Total at grade (L1 + L2)	488.4	603.3	790.9	931.6	1,216
Main exit E1 >= 1/2 (1030.2)	244.2	301.6	395.5	465.8	512
Lose E1: others >= 50% (1005.5)	244.2	301.6	395.5	465.8	704

All four floor cases PASS (floor loads 329, 1,095, 2,346, 3,284; building 3,256, 4,022, 5,273, 6,211). Design case chairs: building 5,273, E1 needs 395.5 vs 512. Stair discharge doors X4 / X6 / X9 / X10 carry the L2 worst case first: 1,460 / 4 x 0.15 = 54.8 in each of 64. Sensitivity, the drawn 120 x 144 ft floor (17,280 SF, Rev D used 114 x 144): Chairs 7 net 2,469 on the floor, building 5,396, E1 404.7 vs 512, total 809.4 vs 1,216; Standing 5 net 3,456 on the floor, building 6,383, E1 478.7 vs 512, total 957.4 vs 1,216 — still PASS.

4 LEVEL 2 STAIR CAPACITY (0.2 in / occupant)

4 stairs x 76 in = 304 in. Worst 1,460 x 0.2 = 292.0 in -> spare 12.0 in (Rev D: 292.4, spare 11.6). Lose one stair: 228 in >= 146.0 in (1005.5). PASS as calculated. This holds only if the ST-2 stair really is 76 in: see finding F1.

5 EXIT LOCATIONS (each tested against the Rev K outline)

EXIT	POSITION	NOTE	
X1	W wall (0.0, 50.0)	team assembly; clear of ST-4 (top y 44.1)	OK
X10	W wall (0.0, 32.0)	ST-4 discharge, door centred on the stair	OK
X9	S wall (183.7, 0.0)	ST-3 discharge, door centred on the stair	OK
X6	tower east face (188.0, 257.7)	ST-2 discharge, opens to open ground NE	OK
X2	W wall (0.0, 98.0)	girls locker (unchanged)	OK
X3	W wall (0.0, 182.0)	boys locker (unchanged)	OK
X4	N wall (42.7, 252.0)	ST-1 discharge (unchanged)	OK
X5	N wall (127.8, 252.0)	EXIT (N); bay clear (x 136-166) (unchanged)	OK
X8	E wall (210.0, 34.0)	east concourse (unchanged)	OK
X7 / X11	bump-out E (unchanged)	EXIT (E) passage / core-2 corridor (D-072)	OK
E1	S wall x 87.25-138.75	8-pair bank (unchanged)	OK

Test: door centre on the envelope line, not in a corner arc, not under the annex or bay, and (for stair discharge) centred on its stair. The outline passes at all points; whether a door opens onto usable ground is a parcel question (D-006).

6 TRAVEL DISTANCE + STAIR SEPARATION

- Level 2 exit access (T1017.2, A sprinklered, 250 ft): worst point on the loop re-measured on Rev G, centreline walk to the nearest stair door + 15 ft seat access (ASSUMED) = 162 ft near (206, 142), the east leg mid-point, longer than Rev D's ≈ 141 ft because ST-2 moved 20 ft west and the method differs (not like-for-like). PASS (< 250). Level 1 floor-to-X7 ≈ 133 ft and core-2 common path (X11) unchanged: the moved exits are on the west and south walls and the tower.
- Stair separation (IBC 2021 1007.1.1, 1/3 of the building diagonal when sprinklered; section not re-read this session): diagonal 328 ft -> 1/3 = 109 ft. Door-to-door: ST-1-ST-2 136, ST-1-ST-3 268, ST-1-ST-4 211, ST-2-ST-3 246, ST-2-ST-4 284, ST-3-ST-4 178 ft. Shortest 136 ft: PASS. (Rev D noted separation not checked.)

7 FINDINGS

F1 FAIL (geometry): the ST-2 tower is too small for the stair. The D-061 / D-052 stair is two 76 in flights side by side, 12.67 x 24.08 ft (31 risers); its diagonal is 27.2 ft but the tower is 20 ft across (inside less wall, about 19). In a 20 ft circle a 12.67 ft wide stair has room for 15.5 ft of length, not 24.08: 8.6 ft short. The Rev K sheet compared floor area (314 vs 305 SF), which does not show this. A helical 76 in stair in 20 ft leaves a 7.3 ft well; curved / helical egress-stair radius and tread rules (IBC 1011) were not checked and a well that small is unlikely to meet them. The capacity in sections 3-4 therefore does NOT yet hold for ST-2.

RECOMMENDATION: make the tower a 29 ft drum (inside ≈ 27.7 ft, holds the 27.2 ft diagonal), centre (182, 262.5), west edge x 167.5, still wrapped by the outline. Effect: L1 and L2 each +341 SF (total +682 GSF on top of 86,212), bay moves 2 ft west to x 134-164 (X5 passage x 123.85-131.85 stays clear; bay to annex gap becomes 12 ft), strip ends about x 167. Not drawn here: it needs A-101 Rev L / A-102 Rev H and your yes (D-081 on approval).

F2 A 6 ft strip landing at the tower door against a 76 in stair needs the architect (IBC 1010 not checked). It improves if the tower grows (F1): the door can face the strip at a wider point.

F3 Level 1 area correction: 50 SF of MECH (N) lies under the tower, so Rev K's room table over-counts MECH (N); the envelope total (52,840) is unaffected.

8 OPEN ITEMS FOR THE ARCHITECT / AHJ

- Everything in Rev D section 12 still stands: editions + AHJ (D-008), construction type, smoke-protected seating, tier structure 1'-6", door widths, seat type.
- ST-2 stair type and size (F1); structure of the curved shell and a 29 ft drum; P2-G-003 Rev L areas.
- SRM / Summertown Metals roles not agreed; descriptive names only (D-042).

Sources: P2-G-002 Rev D; params/phase2_plan_rev_i.yaml, phase2_plan_rev_k.yaml, phase2_plan_rev_g_l2.yaml; IBC 2021 T1004.5, 1005.3, 1005.5, 1030.2-1030.3, T1017.2 (as Rev D); D-052, D-054, D-061, D-077..D-079.

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	CODE ANALYSIS EXIT RECHECK	NTS	October 2026	E	Drawn by Claude (AI) for Shane Brazelton	P2-G-002